

XRPL Validator — Daily Engineering Report • 2026-07-04

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 5 commits (xrpl-validator 5)

Engineering changes

- **feat(node): S1/S2 peer signature verification — proposals + validations (observe-mode)** (03af266 , xrpl-validator) Closes SECURITY 7.3/7.4 measurement gap: TMProposeSet signatures verified over sha512Half('PRP\0' || be32(proposeSeq) || be32(closeTime) || prevLedger || txSetHash) (secp256k1 digest-verify, ed25519 counted unsupported per rippled verifyDigest); STValidation blobs field-walked, Signature stripped, 'VAL\0' preimage verified with key-type dispatch. Counters on /metrics (xrpl_proposal|validation_sig_*_total); ...

Docs and operations

- **chore(node): ffi_test — match apply_ledger_in_order 4 new optional log/expectation params** (551e804 , xrpl-validator)
- **merge: feat/state-rocks-miss-buckets (the validator host tip) into main (state-rocks miss instrumentation, ffi_test signature chore)** (3986917 , xrpl-validator)
- **merge: fix/succ-exclusive-bound into main (6th-halt NFTokenPage fix: succ open interval)** (f995252 , xrpl-validator)
- **chore(ffi): SHIM_VERSION 0.3.0 — 3.2.0 shim port; docs: STATE-2026-07 current-truth (M1 sign-off + stale-claim corrections)** (328f089 , xrpl-validator)

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle Auto-Settlement for Cash Options** (amm-defi) Extend XLS-62's OptionCreate to accept a tfCashSettled flag plus sfSettlementOracleID (a PriceOracle keylet reference); the writer's collateral is loc
- **Native Structured Metadata for Tokens** (tokens) Add a new owner-reserved ledger object ltTOKENMETADATA at keylet::tokenMetadata(issuer, currency-or-MPTIssuanceID), created and updated by a new issue

- **Revocable On-Ledger Attestation Registry** (audit) Add a new owner-reserved ledger object `ltATTESTATION` at `keylet::attestation(attester, sfAttestationID)`, where `sfAttestationID` is an attester-chosen `la`

Appendix — commits

Time	Repo	Commit	Subject
12:02	xrpl-validator	551e804	chore(node): ffi_test — match apply_ledger_in_order 4 new optional log/expectation params
12:04	xrpl-validator	3986917	merge: feat/state-rocks-miss-buckets (the validator host tip) into main (state-rocks miss instrumentation, ffi_test signature chore)
12:04	xrpl-validator	f995252	merge: fix/succ-exclusive-bound into main (6th-halt NFTokenPage fix: succ open interval)
12:09	xrpl-validator	328f089	chore(ffi): SHIM_VERSION 0.3.0 — 3.2.0 shim port; docs: STATE-2026-07 current-truth (M1 sign-off + stale-claim corrections)
12:22	xrpl-validator	03af266	feat(node): S1/S2 peer signature verification — proposals + validations (observe-mode)